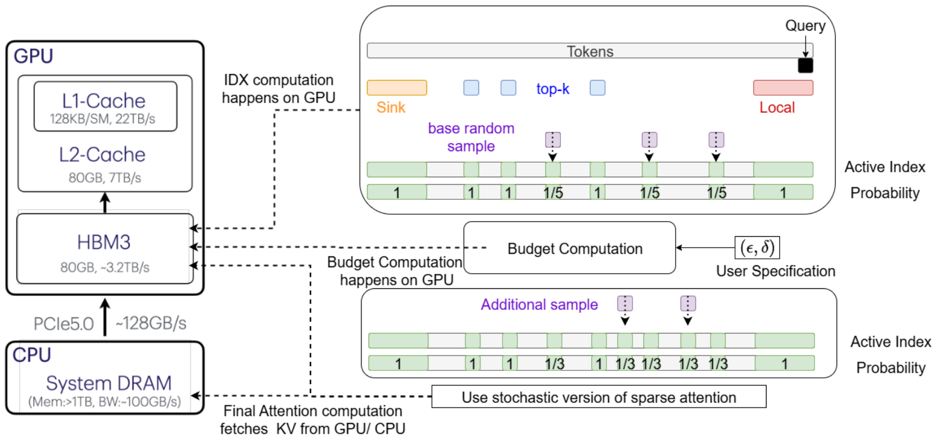

Figure 3: vAttention composes, sink, sliding window, and approximate top-k based attention along with random sampling based selection whose budget is governed by an adaptive sampling module which ensures user specified (ϵ, δ) guarantees hold for each attention head every layer. The index computation and budget computation occur entirely on the GPU, and the final attention computation can retrieve the KV cache from either the GPU/CPU, depending on its location.

of size equal to the budget, and applies sampling-based attention computation. **MagicPig:** Does LSH-based index retrieval followed by sampling estimation. If more tokens are retrieved than the budget allows, a subset of size equal to the budget is randomly selected; otherwise, all retrieved tokens are used. We plot the configuration among $(k = \{2, 4, 8\} \times L = \{8, 16, 32, 64, 128\})$ that yields the best errors for a particular sparsity. We use a sample from the GSM-Infinite (Zhou et al., 2025) dataset of length $25K$ and use the last 128 queries for attention computation.

As shown in Figure 2, the quality–efficiency tradeoff depends on the distribution of attention scores, and no baseline is universally superior. We observe that the ordering is inconsistent—for a given query across heads, or for a given head across queries—highlighting the need for dynamic behavior per head per query. Moreover, when attention scores are sharply distributed, oracle top-k provides a better tradeoff. In contrast, random sampling performs better in the presence of a long tail (looking at the distribution of attention scores sorted in descending order). MagicPig, though based on importance sampling, also fails to outperform other methods consistently. Drawing inspiration from these observations, we propose to combine top-k and sampling methods. As a representative, we use **oracle-top + random-sample:** using half the budget for oracle-top and the other half for sample. We find that this combination consistently yields superior results in all three cases. This hybrid strategy serves as a simplification of our proposed vAttention method.

4 vATTENTION: VERIFIED SPARSE ATTENTION

vAttention breaks down the problem of attention approximation into two parts: (1) identifying outlier or heavy-hitter tokens, and (2) approximating the residual long tail of tokens with similar attention scores using uniform random sampling. The key idea is two fold. First, if the heavy tokens are correctly identified, the contribution of the residual tail can be approximated with a small sample. Second, the convergence properties of uniform sampling can be leveraged to provide guarantees on approximation errors. We describe the exact algorithm and its mathematical foundations below.

Let the KV cache for a given attention head be denoted by $K, V \in \mathbb{R}^{n \times d}$ and the query by $q \in \mathbb{R}^{1 \times d}$. The index selection procedure in vAttention is illustrated in Figure 3. vAttention selects a mixture of deterministic and stochastic indices, and is parameterized by $f_s, f_l, f_t, f_b, \epsilon, \delta \in (0, 1)$. For the deterministic component, vAttention includes sink indices $\mathcal{I}_s, |\mathcal{I}_s| = f_s n$, local window indices $\mathcal{I}_l, |\mathcal{I}_l| = f_l n$, and predicted top-$k$ tokens $\mathcal{I}_t, |\mathcal{I}_t| = f_t n$. These correspond to the most salient tokens expected to dominate the attention distribution, i.e., those with high attention scores. vAttention can be composed with any off-the-shelf approximate top-k method. Incorporating such "heavy hitters" is crucial for strong performance, since the stochastic component—uniformly sampled indices—can only approximate the residual attention accurately with small sample when the remaining distribution